

Semi-prime factorization – Analysis and improvements on Fermat's Factorization method

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Version 4.4

Foreword

This is a work in progress. I need to add a lot more information. However, in this iteration, I hope to compress everything down to as few pages as possible.

No AI was used for this research project. Most of the foundation I uncovered back in 2024 already, but as I had only been studying math for a year, it took me a few additional years to reach the maturity required to make all the right connections. I can however backtrack and explain how I uncovered everything that is described here, from the ground up, with pen and paper, all the way back in 2023 when I started this project.

To do: Add references and closing thoughts and clean everything up.

Chapter I – Fermat's Factorization method

In its most basic form the procedure for Fermat's Factorization method is as follows (where N is the semi-prime being factored).

1. Calculate $\sqrt{N}$ in $\mathbb{Z}$
2. Starting from $x = \sqrt{N}$, calculate $f(x) = x^2 - N$ in a loop, incrementing x .
3. If $f(x)$ evaluates to a square, y^2 , calculate the greatest common divisor (GCD) on the difference of x and y .

Taking $N = 4387$ as an example:

Step 1 (Calculate the +/- square root of N):

$$\sqrt{4387} = 66 \text{ in } \mathbb{Z}$$

Step 2 (Starting from the square root calculate $f(x) = x^2 - N$:

$66^2 - 4387 = -31$ (not a square)
 $67^2 - 4387 = 102$ (not a square)
 $68^2 - 4387 = 237$ (not a square)
 $69^2 - 4387 = 374$ (not a square)
 $70^2 - 4387 = 513$ (not a square)
 $71^2 - 4387 = 654$ (not a square)
 $72^2 - 4387 = 797$ (not a square)
 $73^2 - 4387 = 942$ (not a square)
 $74^2 - 4387 = 1089 = 33^2$ (square)

Step 3 (If the result is also a square, calculate the GCD on the difference.):

$$74^2 = 33^2 \text{ mod } 4387$$

$$\text{GCD}(74 + 33, 4387) = 107$$
$$\text{GCD}(74 - 33, 4387) = 41$$

Both the Quadratic Sieve and Number Field Sieve algorithms (to do: insert references) are expansions on this method.

Chapter II – Analysis

a. Binomial expansion

As seen in chapter 1 the goal is the find the following numbers that we can use to take the GCD with:

$$\text{GCD}(x + y, N) = q$$
$$\text{GCD}(x - y, N) = p$$

x and y can thus be rewritten as the differences of the factors of N , $p + q$ or $p - q$.
In addition, many more such solutions exist created by $(ap + bq)/2$ and $(ap - bq)/2$ where a and b are multipliers.

If $a = 13$, $b = 5$ and $N = 4387$ (and thus factors $p = 41$ and $q = 107$)

We get:

$$(13 \times 41 + 5 \times 107)/2 = 534$$
$$(13 \times 41 - 5 \times 107)/2 = -1$$

And we can see this creates the following expression when we square 534 modulo N (4387 here):

$$534^2 - 4387 \times 13 \times 5 = 1^2$$

This behavior can be explained using binomial expansion.

I will shorten binomial expansion as $BE(x - y)^e$.
For example $BE(x - 74)^2 = x^2 - 148x + 5476$.

Substituting the binomial term x , with for example 33, we get:

$$BE(33 - 74)^1 = 33 - 74 = -41$$
$$BE(33 - 74)^2 = 33^2 - 148 \times 33 + 5476 = -41^2$$
$$BE(33 - 74)^3 = 33^3 - 222 \times 33^2 + 16428 \times 33 - 405224 = -41^3$$

$$BE(33 - 74)^4 = 33^4 - 296 \times 33^3 + 32856 \times 33^2 - 1620896 \times 33 + 29986576 = -41^4$$

Conversely we also see the following occurring if we substitute in the factor as root instead (or swapping a binomial term with the binomial value):

$$\begin{aligned} BE(41 - 74)^1 &= 41 - 74 = -33 \\ BE(41 - 74)^2 &= 41^2 - 148 \times 41 + 5476 = -33^2 \\ BE(41 - 74)^3 &= 41^3 - 222 \times 41^2 + 16428 \times 41 - 405224 = -33^3 \\ BE(41 - 74)^4 &= 41^4 - 296 \times 41^3 + 32856 \times 41^2 - 1620896 \times 41 + 29986576 = -33^4 \end{aligned}$$

In addition, if we have even exponents we can substitute the constant by a multiple of N and generate a polynomial p to solve for 0.

$$\begin{aligned} BE(41 - 74)^2 &= 41^2 - 148 \times 41 + 5476 \rightarrow p = 41^2 - 148 \times 41 + 4387 = 0 \\ BE(41 - 74)^4 &= 41^4 - 296 \times 41^3 + 32856 \times 41^2 - 1620896 \times 41 + 29986576 \rightarrow p = 41^4 - 296 \times 41^3 + 32856 \times 41^2 - 1620896 \times 41 + 4387 \times 6565 = 0 \end{aligned}$$

b. The discriminant

Of-course we cannot blindly perform binomial expansion on random numbers. We need some type of measure indicating us how close or far away we are from a solution.

Take for example the following binomial expansion $BE(x - 74)^2 = x^2 - 148x + 5476$ and on this quadratic polynomial p , we substitute the constant with N (for example $N = 4387$):

$$p = x^2 - 148x + 4387$$

If we now calculate the discriminant of p , $DISC(p) = 148^2 - 4 \times 4387 = 66^2$, we see that the result is also square. And we now have the square relation $148^2 = 66^2 \pmod{4387}$, whose $GCD(148 - 66, 4387) = 41$.

An observation:

From this we can make the following generalization that we can derive the factorization of a semi-prime N , if we find a quadratic polynomial with a constant set to N , or a multiple thereof, and whose discriminant yields a square. And from this we can also assume that the discriminant would yield a quadratic residue modulo any prime.

c. Using binomial expansions

The below examples assume $N = 4387$.

Previously we had the following binomial b :

$$b = 33 - 74 = -41$$

Next performing binomial expansion to degree 2, if we then substitute the constant by N to create polynomial p , and swap a binomial term with the binomial value (such that $b = 41 - 74$ instead), if p evaluates to 0, we have a valid solution.

$$BE(41 - 74)^2 = 41^2 - 148 \times 41 + 5476 = 33^2 \rightarrow p = 41^2 - 148 \times 41 + 4387 = 0$$

For binomial term $y = 74$, only binomial term $x = 33$ will yield a correct binomial value using this setup. Since 74^2 and 33^2 are congruent mod 4387. Further more calculating the discriminant on polynomial p will yield a square multiple of binomial term x , for example, $DISC(p) = (2 \times x)^2$ if we have a correct solution. See previous sub-chapter.

The main issue with this setup when performing binomial expansion using $BE(x - y)^e$, is that when expanding to even exponents, we need to know which multiple of N to substitute with. Because this will determine whether or not a root solution exists, derived from the binomial terms, that will evaluate the Quadratic to 0. Hence using this oracle setup we can restate the factorization problem as finding a multiple of N to substitute the constant value in $BE(x - y)^e$, where e is even, with. Luckily for us, we can calculate the possible multiples of N modulo m (where m is a product of primes). And this is what I propose as an improvement in the next chapter (Chapter 3 – Improvements).

d. An extended example

Let $N = 4387$

Let binomial $b = x - 534$.

$$\text{Hence } BE(x - 534)^2 = x^2 - 1068x + 534^2$$

We now need to substitute the constant in the resulting polynomial with a multiple of N . For example let multiple $k = 1$.

$$k = 1 \rightarrow p = x^2 - 1068x + 4387$$

$DISC(p)$ is not a square in $\mathbb{Z}$ hence we know $k = 1$ cannot be a valid solution.

However if $k = 65$:

$$k = 65 \rightarrow p = x^2 - 1068x + 4387 \times 65 \text{ and } DISC(p) = 2^2$$

The discriminant is defined as $DISC(p) = (2 \times x)^2$, where x is the binomial term, we can now substitute this into b .

$b(1) = 1 - 534 = -533$ and $p(533) = 0$. Since p evaluates to 0 in $\mathbb{Z}$ we know we have a correct solution and we can take the gcd using our binomial terms:

$$\begin{aligned} GCD(1 + 534, 4387) &= 107 \\ GCD(534 - 1, 4387) &= 41 \end{aligned}$$

In the next chapter we will see how to find these solutions in $\mathbb{Z}/p\mathbb{Z}$

Chapter 3 – Improvements

a. Calculating polynomial residues in $\mathbb{Z}/p\mathbb{Z}$

Let us restrict ourselves to binomial expansions to the second degree.

The way we calculate polynomial residues is that we must consider all possible coefficients in $\mathbb{Z}/p\mathbb{Z}$ whose discriminant evaluates to a quadratic residue.

Here we calculate all possible residues for the quadratic polynomial $ax^2 - bx + Nk$ where $N = 4387$ (you can also calculate them for $ax^2 + bx - Nk$ instead, it just changes the side we are calculating in the expression $x^2 = y^2 \pmod{N}$). We will ignore the cases where the leading coefficient is 0, as this would downgrade the degree of our polynomial. And in addition we will also ignore the case where $k = 0$ for the sake of brevity.

Mod 3

$$k = 1, a = 1 \text{ and } b = -1 \rightarrow x = \{2\}, k = 1, a = 1 \text{ and } b = -2 \rightarrow x = \{1\}, k = 1, a = 2 \text{ and } b = 0 \rightarrow x = \{1, 2\}$$

$$k = 2, a = 1 \text{ and } b = 0 \rightarrow x = \{2, 1\}, k = 2, a = 2 \text{ and } b = -1 \rightarrow x = \{1\}, k = 2, a = 2 \text{ and } b = -2 \rightarrow x = \{2\}$$

Mod 5

$$k = 1, a = 1 \text{ and } b = -2 \rightarrow x = \{4, 3\}, k = 1, a = 1 \text{ and } b = -3 \rightarrow x = \{2, 1\}, k = 1, a = 2 \text{ and } b = 0 \rightarrow x = \{3, 2\}, \\ k = 1, a = 2 \text{ and } b = -1 \rightarrow x = \{4\}, k = 1, a = 2 \text{ and } b = -4 \rightarrow x = \{1\}, k = 1, a = 3 \text{ and } b = 0 \rightarrow x = \{1, 4\}, \\ k = 1, a = 3 \text{ and } b = -2 \rightarrow x = \{2\}, k = 1, a = 3 \text{ and } b = -3 \rightarrow x = \{3\}, k = 1, a = 4 \text{ and } b = -1 \rightarrow x = \{3, 1\}, \\ k = 1, a = 4 \text{ and } b = -4 \rightarrow x = \{4, 2\}$$

$$k = 2, a = 1 \text{ and } b = 0 \rightarrow x = \{4, 1\}, k = 2, a = 1 \text{ and } b = -1 \rightarrow x = \{3\}, k = 2, a = 1 \text{ and } b = -4 \rightarrow x = \{2\}, \\ k = 2, a = 2 \text{ and } b = -1 \rightarrow x = \{2, 1\}, k = 2, a = 2 \text{ and } b = -4 \rightarrow x = \{3, 4\}, k = 2, a = 3 \text{ and } b = -2 \rightarrow x = \{3, 1\}, \\ k = 2, a = 3 \text{ and } b = -3 \rightarrow x = \{4, 2\}, k = 2, a = 4 \text{ and } b = 0 \rightarrow x = \{2, 3\}, k = 2, a = 4 \text{ and } b = -2 \rightarrow x = \{4\}, \\ k = 2, a = 4 \text{ and } b = -3 \rightarrow x = \{1\}$$

$$k = 3, a = 1 \text{ and } b = 0 \rightarrow x = \{3, 2\}, k = 3, a = 1 \text{ and } b = -2 \rightarrow x = \{1\}, k = 3, a = 1 \text{ and } b = -3 \rightarrow x = \{4\}, \\ k = 3, a = 2 \text{ and } b = -2 \rightarrow x = \{2, 4\}, k = 3, a = 2 \text{ and } b = -3 \rightarrow x = \{1, 3\}, k = 3, a = 3 \text{ and } b = -1 \rightarrow x = \{3, 4\}, \\ k = 3, a = 3 \text{ and } b = -4 \rightarrow x = \{2, 1\}, k = 3, a = 4 \text{ and } b = 0 \rightarrow x = \{4, 1\}, k = 3, a = 4 \text{ and } b = -1 \rightarrow x = \{2\}, \\ k = 3, a = 4 \text{ and } b = -4 \rightarrow x = \{3\}$$

$$k = 4, a = 1 \text{ and } b = -1 \rightarrow x = \{2, 4\}, k = 4, a = 1 \text{ and } b = -4 \rightarrow x = \{1, 3\}, k = 4, a = 2 \text{ and } b = 0 \rightarrow x = \{4, 1\}, \\ k = 4, a = 2 \text{ and } b = -2 \rightarrow x = \{3\}, k = 4, a = 2 \text{ and } b = -3 \rightarrow x = \{2\}, k = 4, a = 3 \text{ and } b = 0 \rightarrow x = \{2, 3\}, \\ k = 4, a = 3 \text{ and } b = -1 \rightarrow x = \{1\}, k = 4, a = 3 \text{ and } b = -4 \rightarrow x = \{4\}, k = 4, a = 4 \text{ and } b = -2 \rightarrow x = \{1, 2\}, \\ k = 4, a = 4 \text{ and } b = -3 \rightarrow x = \{3, 4\}$$

The problem is, that we can Chinese remainder these together to modulo 15, but then the solution set of possible residues grows quickly (the product of the size of both sets of solutions). So we need to come up with a smart way to generate solutions with this setup.

b. Singular quadratic roots vs non-singular quadratic roots

In the above residue mapping, we will observe that sometimes a set of coefficients will have a single root solution, and sometimes it will have two root solutions.

For example, looking in Mod 5 we see the following two coefficient, root pairings:

$$k = 1, a = 4 \text{ and } b = -1 \rightarrow x = \{3, 1\} \text{ Non-singular roots}$$

$$k = 1, a = 3 \text{ and } b = -2 \rightarrow x = \{2\} \text{ Singular root}$$

Singular roots indicate that the corresponding binomial term has prime p (5 here) as divisor and that the discriminant is also divided by prime p .

We have calculated the above residues using polynomials of the shape $ax^2 - bx + Nk$.

If we calculate the derivative in $\mathbb{Z}/5\mathbb{Z}$ we get the residue for linear coefficient b in $\mathbb{Z}/5\mathbb{Z}$ corresponding to a quadratic with the signs flipped $aax^2 + bx - Nk$ (the other side of the expression $x^2 = y^2 \pmod{N}$ that we are trying to solve for).

Calculating the derivative for coefficients with a singular root in $\mathbb{Z}/p\mathbb{Z}$ gives us a derivative that is 0 in $\mathbb{Z}/p\mathbb{Z}$. Since the derivative calculates the linear coefficient for the other side of the expression $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$, and since the linear coefficient is constructed by multiplying the binomial term by 2, any divisors of the linear coefficient are hence also divisors of the binomial term.

For example using: $k = 1, a = 3 \text{ and } b = -2 \rightarrow x = \{2\}$ we construct polynomial $p = 3 \times 2^2 - 2 \times 2 + 2$ in $\mathbb{Z}/5\mathbb{Z}$.

And we see that the derivative $p' = 2 \times 3 \times 2 - 2 = 0 \pmod{5}$ (or in other words 0 in $\mathbb{Z}/5\mathbb{Z}$).

This implies that the binomial term y , in binomial $b = y - x$ equals to 0 in $\mathbb{Z}/5\mathbb{Z}$.

Non-singular roots always yield non-zero residues for the binomial terms in $\mathbb{Z}/p\mathbb{Z}$.

c. p-adic lifting of polynomials in $\mathbb{Z}/p\mathbb{Z}$

Let us say we have the following coefficient and root pairing from our residue mapping:

$$k = 4, a = 3 \text{ and } b = -1 \rightarrow x = \{1\} \text{ in } \mathbb{Z}/5\mathbb{Z} \text{ gives us polynomial } p = 3 \times 1^2 - 1 \times 1 + 4387 \times 4 = 0 \pmod{5}$$

This gives us all of the following possible solutions in $\mathbb{Z}/25\mathbb{Z}$ where also the derivative yields 0 in $\mathbb{Z}/25\mathbb{Z}$, for the sake of brevity we will focus on these cases only:

$$k = 4, a = 3 \text{ and } b = -1 \rightarrow x = \{21\}, k = 4, a = 8 \text{ and } b = -6 \rightarrow x = \{16\}, k = 4, a = 13 \text{ and } b = -11 \rightarrow x = \{11\} \\ k = 4, a = 18 \text{ and } b = -16 \rightarrow x = \{6\}, k = 4, a = 23 \text{ and } b = -21 \rightarrow x = \{1\}$$

$$k = 9, a = 3 \text{ and } b = -11 \rightarrow x = \{6\}, k = 9, a = 8 \text{ and } b = -16 \rightarrow x = \{1\}, k = 9, a = 13 \text{ and } b = -21 \rightarrow x = \{21\} \\ k = 9, a = 18 \text{ and } b = -1 \rightarrow x = \{16\}, k = 9, a = 23 \text{ and } b = -6 \rightarrow x = \{11\}$$

$$k = 14, a = 3 \text{ and } b = -21 \rightarrow x = \{16\}, k = 14, a = 8 \text{ and } b = -1 \rightarrow x = \{11\}, k = 14, a = 13 \text{ and } b = -6 \rightarrow x = \{6\} \\ k = 14, a = 18 \text{ and } b = -11 \rightarrow x = \{1\}, k = 14, a = 23 \text{ and } b = -16 \rightarrow x = \{21\}$$

$k = 19, a = 3$ and $b = -6 \rightarrow x = \{1\}$, $k = 19, a = 8$ and $b = -11 \rightarrow x = \{21\}$, $k = 19, a = 13$ and $b = -16 \rightarrow x = \{16\}$
 $k = 19, a = 18$ and $b = -21 \rightarrow x = \{11\}$, $k = 19, a = 23$ and $b = -1 \rightarrow x = \{6\}$

$k = 24, a = 3$ and $b = -16 \rightarrow x = \{11\}$, $k = 24, a = 8$ and $b = -21 \rightarrow x = \{6\}$, $k = 24, a = 13$ and $b = -1 \rightarrow x = \{1\}$
 $k = 24, a = 18$ and $b = -6 \rightarrow x = \{21\}$, $k = 24, a = 23$ and $b = -11 \rightarrow x = \{16\}$

Each of these solutions map to the same solution in $\mathbb{Z}/5\mathbb{Z}$. We see a total of 25 possible solutions in $\mathbb{Z}/25\mathbb{Z}$ generated from a single solution in $\mathbb{Z}/5\mathbb{Z}$. I refer to p-adic lifting, but usually it is used in the context of lifting roots of polynomials, but in the context that I use this term, it is also used to lift coefficient residues in $\mathbb{Z}/p\mathbb{Z}$ p-adically. So please keep that in mind. I tend to create these abstractions in my head and the names I give these abstract concepts may not be strictly correct in formal mathematical terms. But to me it would seem, that whether we are lifting root residues or coefficient residues, the underlying mechanism are very similar and connected.

D. Converting singular roots to non-singular roots and visa versa

Because we are searching for a two squares such that $x^2 = y^2 \pmod N$, and since both x and y can also be represented as binomials which can be expanded to the second degree, we can substitute the expression $x^2 = y^2 \pmod N$ with the more elaborate: $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$.

This also means that we can convert singular roots to non-singular roots, simply by transferring a root to the other side of the equation in $\mathbb{Z}/p\mathbb{Z}$. As long as either b_0 or b_1 does not have p as divisor.

For example $1x^2 + 66x - 4387 = 0 \pmod{37}$ has a singular root, while the converse $1x^2 - 0x + 4387 = 0 \pmod{37}$ has two roots.

$$b_1 = 66 \rightarrow 1 \times 4^2 + 66 \times 4 - 4387 = 0 \pmod{37}$$

$$b_0 = 0 \rightarrow 1 \times 4^2 - 0 \times 33 + 4387 = 0 \pmod{37}$$

$$b_0 = 0 \rightarrow 1 \times 33^2 - 0 \times 33 + 4387 = 0 \pmod{37}$$

But if we flip the sign on the linear coefficient b_1 , then the singular root becomes the other non-singular root for b_0 (we must also swap a and k , but both are 1 here):

$$b_1 = -66 \rightarrow 1 \times 33^2 - 66 \times 33 - 4387 = 0 \pmod{37}$$

Chapter 4 – Constructing a better Quadratic Sieve variant

Traditional Quadratic Sieve algorithms are mainly bottle-necked by the fact that beyond 110 digits the factor base required for successful factorization becomes too large. This is mainly due to the relation between factor base size and the on average number of B -smooths required to succeed at the linear algebra step. Self Initializing Quadratic Sieve has some improvements, the biggest one being the speed it up gets by guaranteeing the factorization of $\pm$ half of the B -smooth candidates being generated. And it also slightly improves our chances of succeeding at linear algebra earlier, as there will be more overlap between factors. SIQS is somewhat the cutting edge of Quadratic Sieve Variants. So we need to find a novel mechanism that we can exploit to get past this issue.

So far we have covered some of the basics. How these findings relate to the binomial theorem. We can calculate residues and perform p-adic lifting. Gaining an advantage out of this setup is something that has occupied me for a long time.

Very early in my research phase I was able to establish that sieving for $x^2 = y^2 \pmod N$ can also be represented as $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$.

This means that we now have an algebraic expression on both sides. Furthermore, we can construct a two-sided sieve with this, where both $ax^2 - b_0x + Nk$ and $ax^2 + b_1x - Nk$ must yield a B -smooth.

We've also shown how this same setup can be constructed for any other even degree. The same construction can be achieved with 4th degree polynomials on each side, and so on.

What is interesting about a setup like $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$, is that both sides can be reframed as a product of two linear polynomials (since a quadratic is simply a product of two linear polynomials). Hence the greatest factor either the left side or right side can contain, is bounded by the sizes of the polynomial values generated by each individual linear polynomials. For a quadratic, the two linear polynomials can be found as x and $f(x)/x$. For example if both x and $f(x)/x$ are roughly 50 bits each. Then the largest factor of the product cannot exceed 50 bits, even though the product might be 100 bits.

If we go up in degree, and we keep a similar structure where $f(x) = g(fx)$, then the amount of linear polynomials that each polynomial value can be represented as is equal to the degree. Hence further subdividing each side and lowering the bounds for what the largest possible factors can possibly be, assuming each linear polynomial produces a polynomial value of roughly equal size.

To do: finish this